

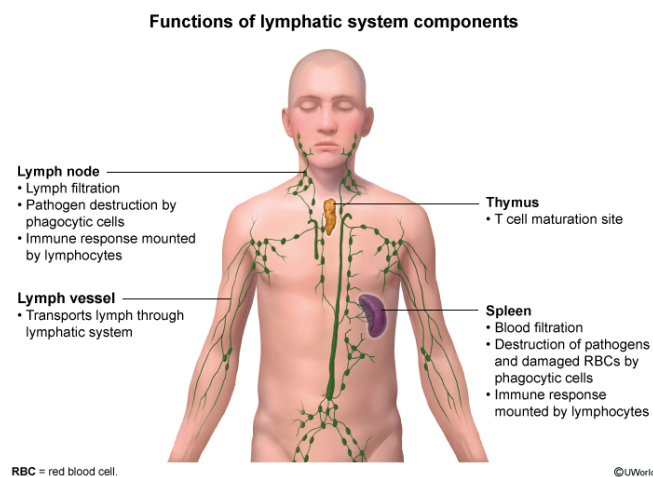
Bacteria are removed from lymph and blood in the:

- ☐ A. spleen and bone marrow, respectively. [6%]
- ✓ ☒ B. lymph nodes and spleen, respectively. [85%]
- ☐ C. bone marrow and lymph nodes, respectively. [2%]
- ☐ D. lymph vessels and thymus, respectively. [5%]

Correct

84% answered correctly

Explanation:



The **lymphatic system** is a network of vessels that collect and transport **lymph**, ultimately draining this fluid into the bloodstream. The lymphatic system is particularly important to the immune response and includes the following **lymphoid tissues** and **organs**, which facilitate **immune cell** proliferation and function:

- The **bone marrow**, a spongy tissue lining the inside of bones, is responsible for the **production** of red blood cells and white blood cells (eg, lymphocytes) from hematopoietic stem cells.
- The **thymus** receives immature T lymphocytes migrating from the bone marrow and facilitates their maturation. During this process, any T lymphocytes that recognize **self-antigens** **die**, preventing the release of lymphocytes able to mount attacks against the body's own tissues. However, T lymphocytes that do *not* bind self-antigens mature and enter the blood and lymph, where they bind foreign antigens and mount **immune responses**.

- **Lymph nodes**, organs clustered along lymph vessels, **filter lymph** as it flows through the lymphatic system. Macrophages housed within lymph nodes identify a variety of pathogens in lymph and destroy them via **phagocytosis**. In addition, **B lymphocytes** and T lymphocytes bind specific foreign antigens in lymph, inducing immune responses against invading pathogens.
- The **spleen** has extensive capillary networks through which **blood is filtered**. White blood cells housed within the spleen recognize and bind foreign antigens in the blood, mounting immune responses against these circulating pathogens. In addition, spleen macrophages remove and destroy old and damaged red blood cells. Platelets and other white blood cells are also stored in the spleen and released into the blood as needed.

Because lymph nodes filter lymph and the spleen filters blood, **bacteria circulating through lymph are removed in the lymph nodes** and **bacteria circulating through blood are removed in the spleen**.

(Choices A and C) Although white blood cells are produced in bone marrow, bone marrow cells do *not* remove bacteria from blood or lymph.

(Choice D) Immune cells circulate throughout the body via lymph vessels; however, pathogens are mainly removed from lymph at lymph nodes (*not* lymph vessels). The thymus is the site of T-cell maturation, but it does *not* filter blood or lymph.

Educational objective:

In the lymph nodes, white blood cells remove and mount immune responses against pathogens from the lymph, whereas in the spleen, white blood cells remove pathogens and damaged or old red blood cells from the blood.

Time spent:0 sec

Subject:

Biology

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System:

Skin and Immune Systems